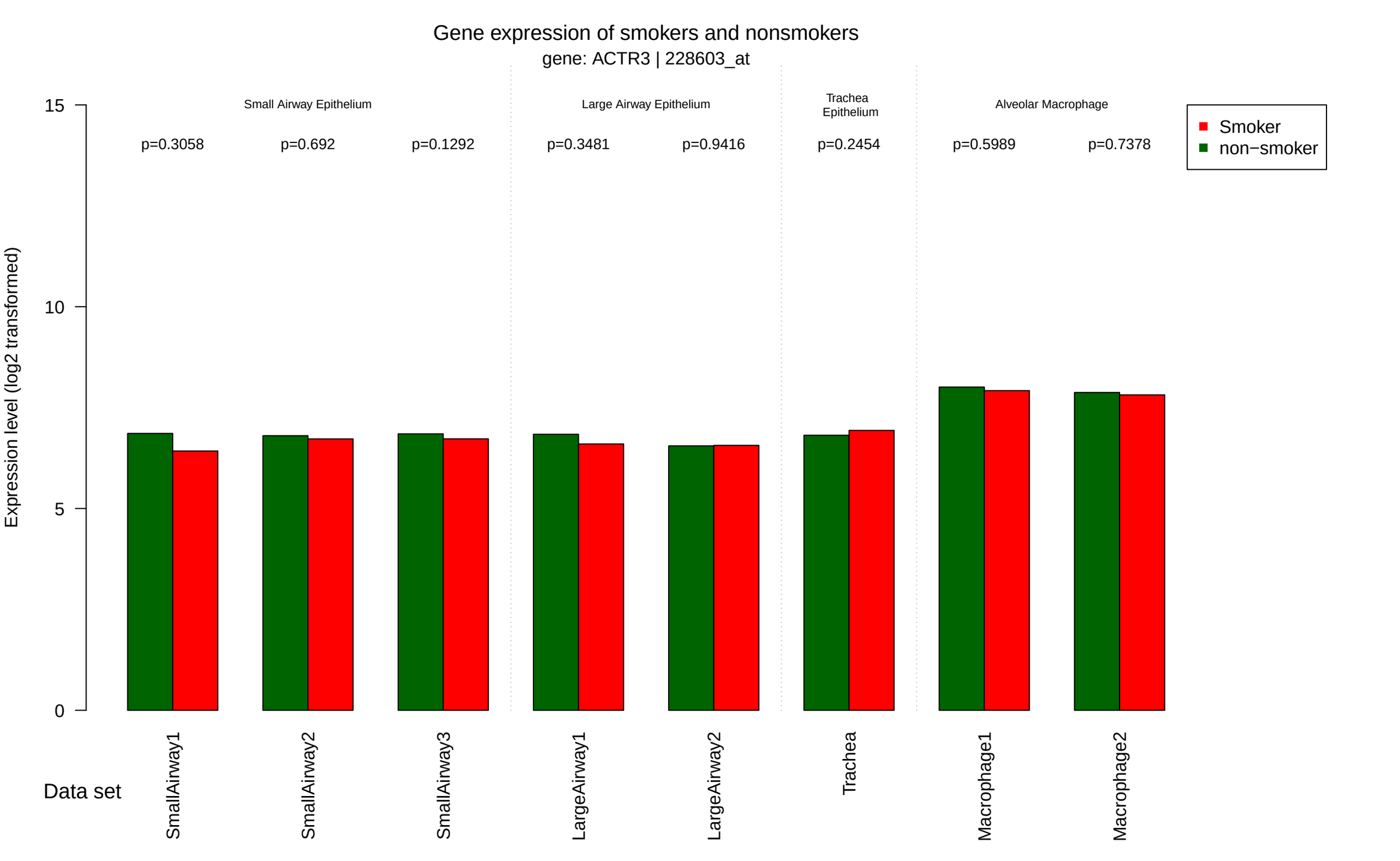
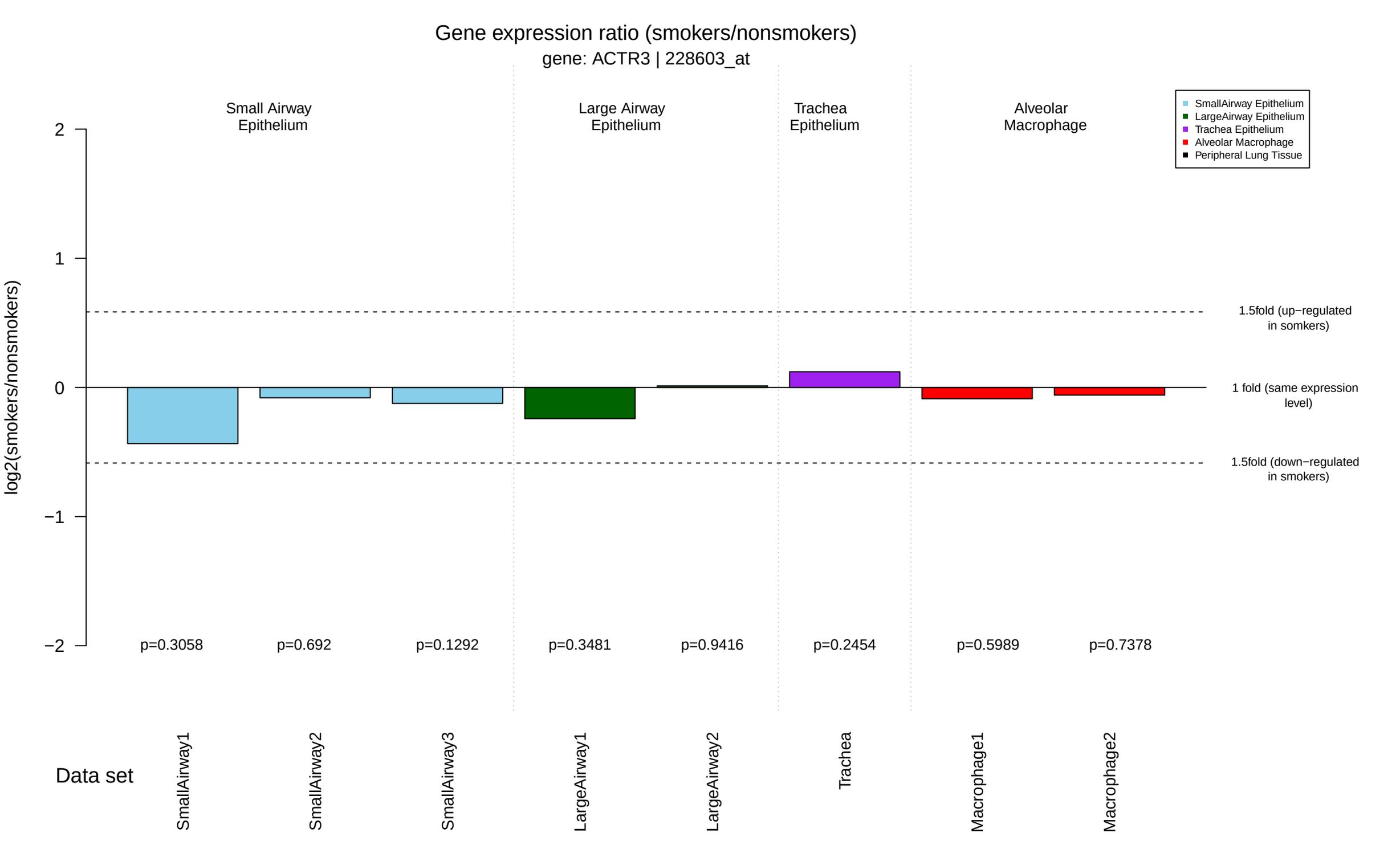
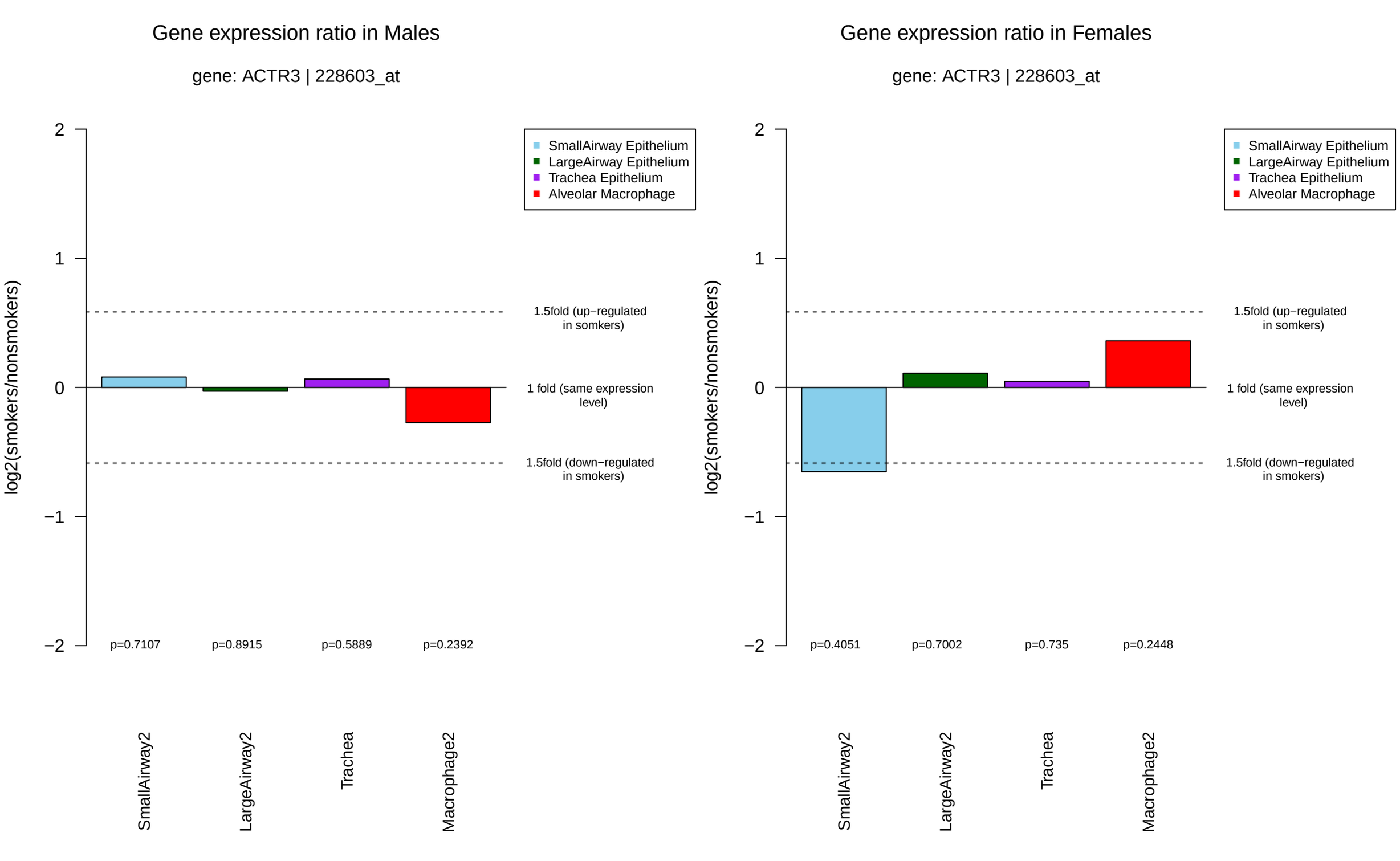
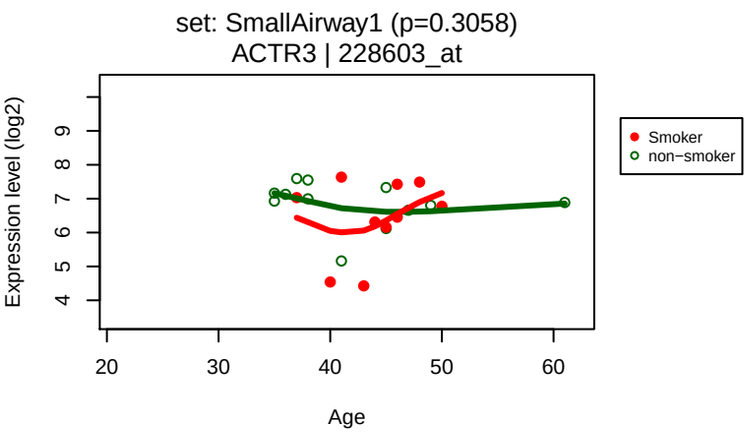
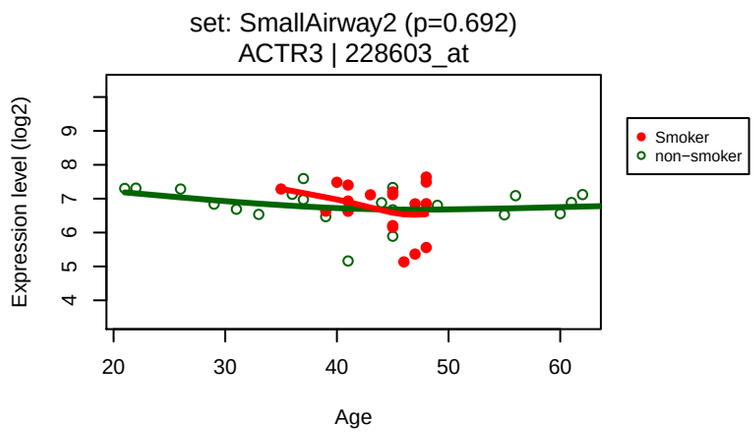
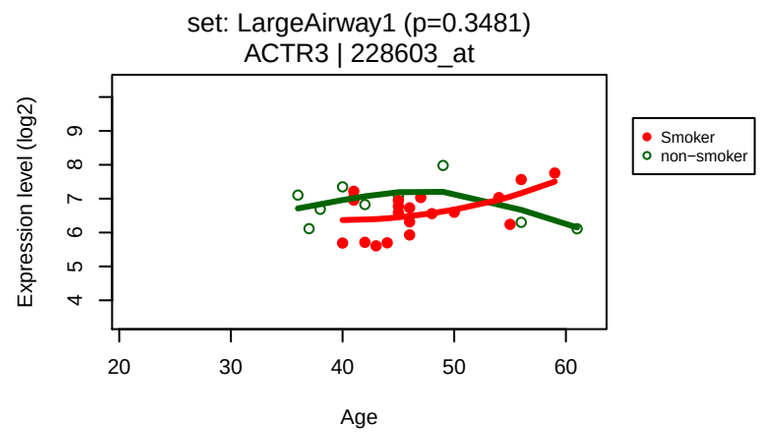

Gene expression of smokers and nonsmokers
gene: ACTR3 | 228603_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3058
p=0.692
p=0.1292
p=0.3481
p=0.9416
p=0.2454
p=0.5989
p=0.7378
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

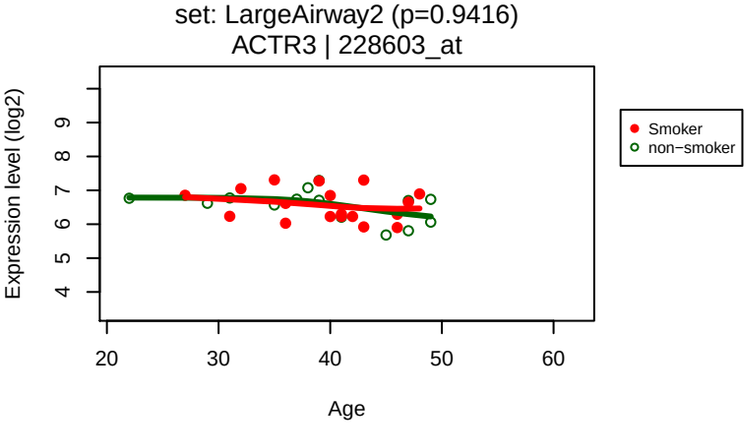

Gene expression ratio (smokers/nonsmokers)
gene: ACTR3 | 228603_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3058
p=0.692
p=0.1292
p=0.3481
p=0.9416
p=0.2454
p=0.5989
p=0.7378
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

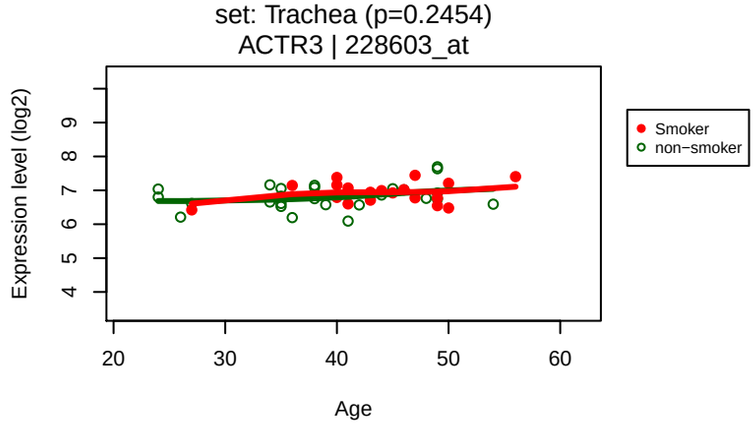

Gene expression ratio in Males
gene: ACTR3 | 228603_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7107
p=0.8915
p=0.5889
p=0.2392
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR3 | 228603_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4051
p=0.7002
p=0.735
p=0.2448
SmallAirway2
LargeAirway2
Trachea
Macrophage2

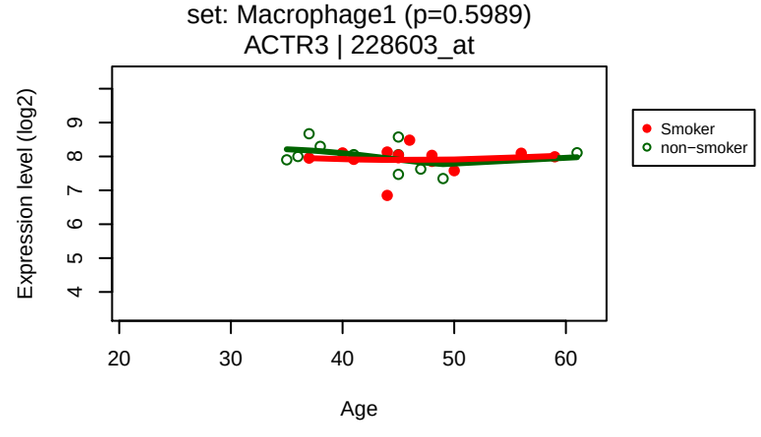

set: SmallAirway1 (p=0.3058)
ACTR3 | 228603_at
Expression level (log2)
Age
Smoker
non−smoker

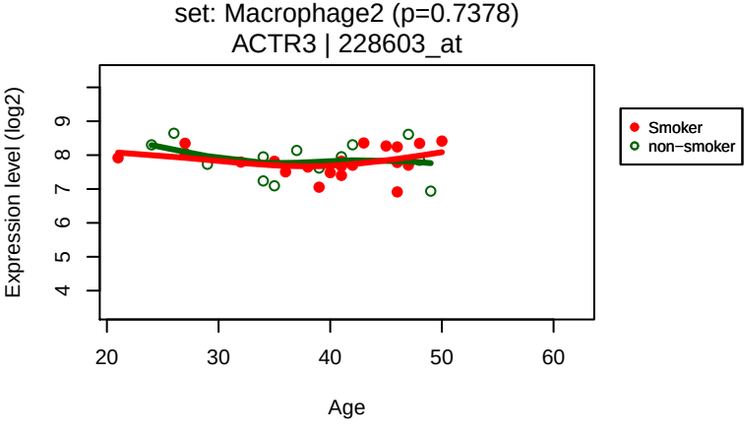

set: SmallAirway2 (p=0.692)
ACTR3 | 228603_at
Expression level (log2)
Age
Smoker
non−smoker

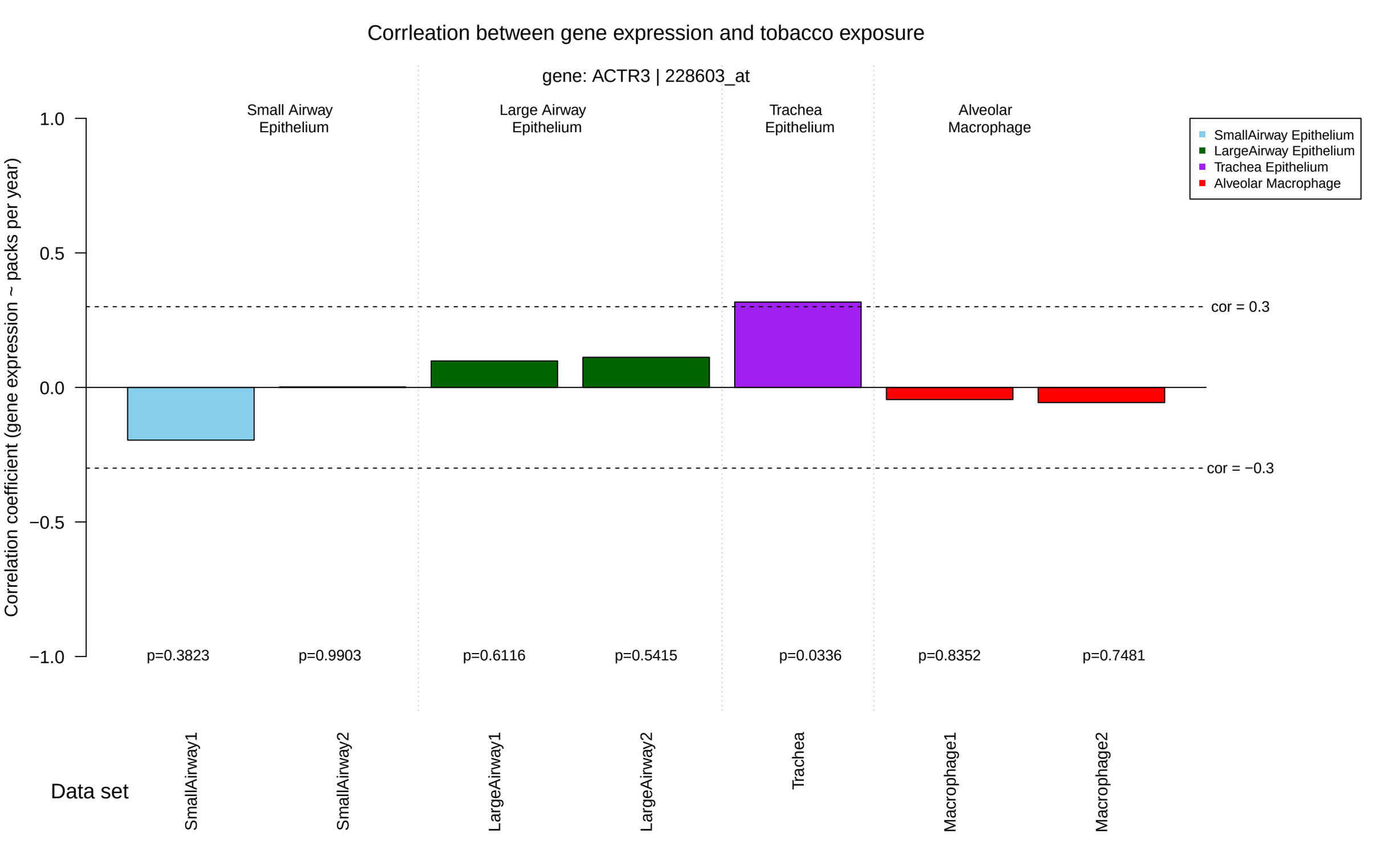

set: LargeAirway1 (p=0.3481)
ACTR3 | 228603_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9416)
ACTR3 | 228603_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2454)
ACTR3 | 228603_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5989)
ACTR3 | 228603_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7378)
ACTR3 | 228603_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR3 | 228603_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3823
p=0.9903
p=0.6116
p=0.5415
p=0.0336
p=0.8352
p=0.7481
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2